



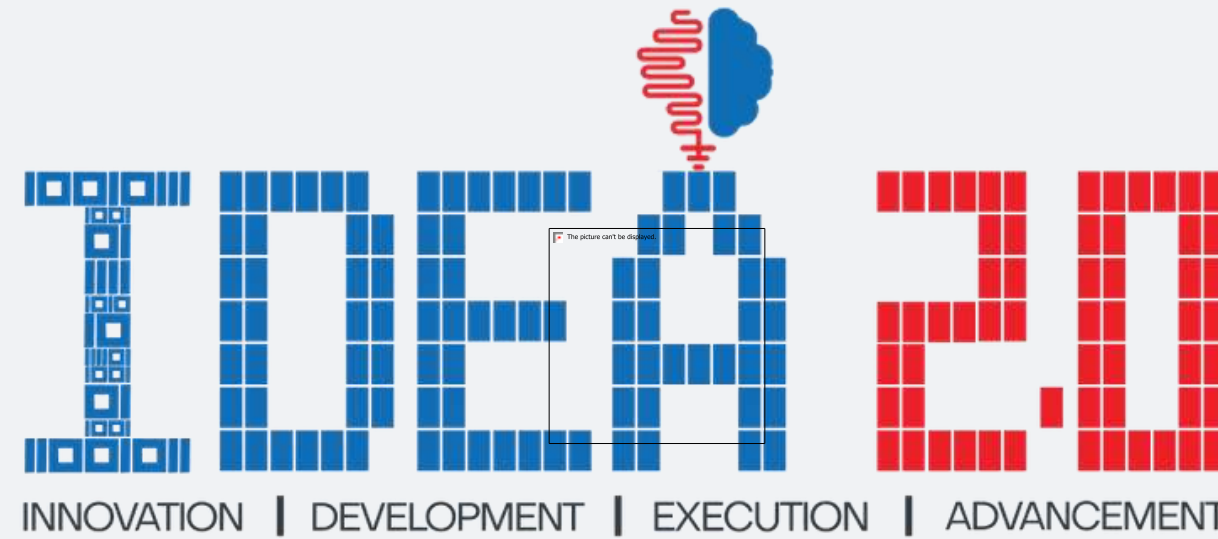
वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



Ai - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME		SIGMOID	
PROBLEM STATEMENT TITLE		AI Driven Early Warning System for Internal & Privileged User Fraud	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Rehaan Ahmad Khan	Male	Machine Learning & Anomaly Detection
Member 2	Krishna Agarwaal	Male	Frontend Development
Member 3	Vishnu Tripathi	Male	Data Engineering
Member 4	Shantanu Singh	Male	Backend Development

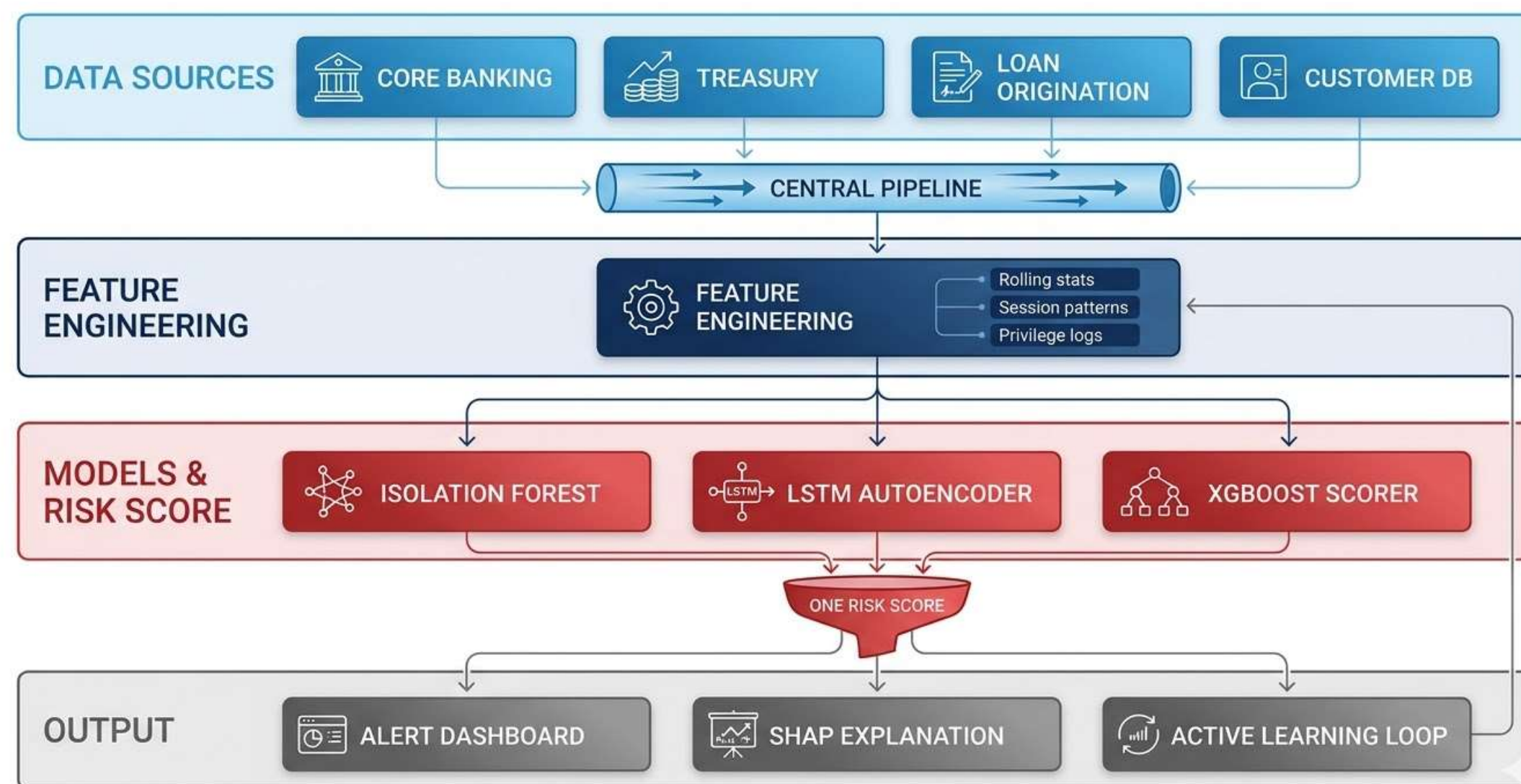
SentinelIQ — AI-Powered Behavioural Early Warning System

Proposed Solution:

- Monitors internal & privileged user behaviour in real time across core banking, treasury, loan origination, and customer databases using an ensemble of ML models
- Builds a dynamic per-user behavioural baseline — learning normal login timing, transaction volume, data access frequency, and privilege usage patterns
- Any deviation (off-hours logins, bulk downloads, unauthorized modifications, privilege escalation) triggers an instant alert with a risk score + SHAP plain-English explanation
- Isolation Forest catches sudden point anomalies, LSTM Autoencoder detects slow behavioural drift over days/weeks, XGBoost combines signals into a single 0–100 composite risk score
- Three-model consensus required before HIGH alert fires — drastically reducing false positive rate
- Active learning loop — investigator labels feed automated weekly retraining, compounding model accuracy over time

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

- **Per-User Dynamic Baselines** — every employee gets their own behavioural fingerprint built on 90-day rolling history, not a global threshold
- **3-Model Ensemble** — Isolation Forest (sudden spikes) + LSTM Autoencoder (slow drift over days) + XGBoost (final composite risk score 0–100)
- **Explainability-First via SHAP** — every alert shows exactly which behaviours drove the score, making it auditable and RBI-compliant
- **Active Learning Loop** — investigator TP/FP labels trigger automated weekly retraining via SQLite-backed feedback pipeline
- **Human-in-the-Loop Architecture** — no automated account action taken; investigation-only by design, fully aligned with RBI operational guidelines



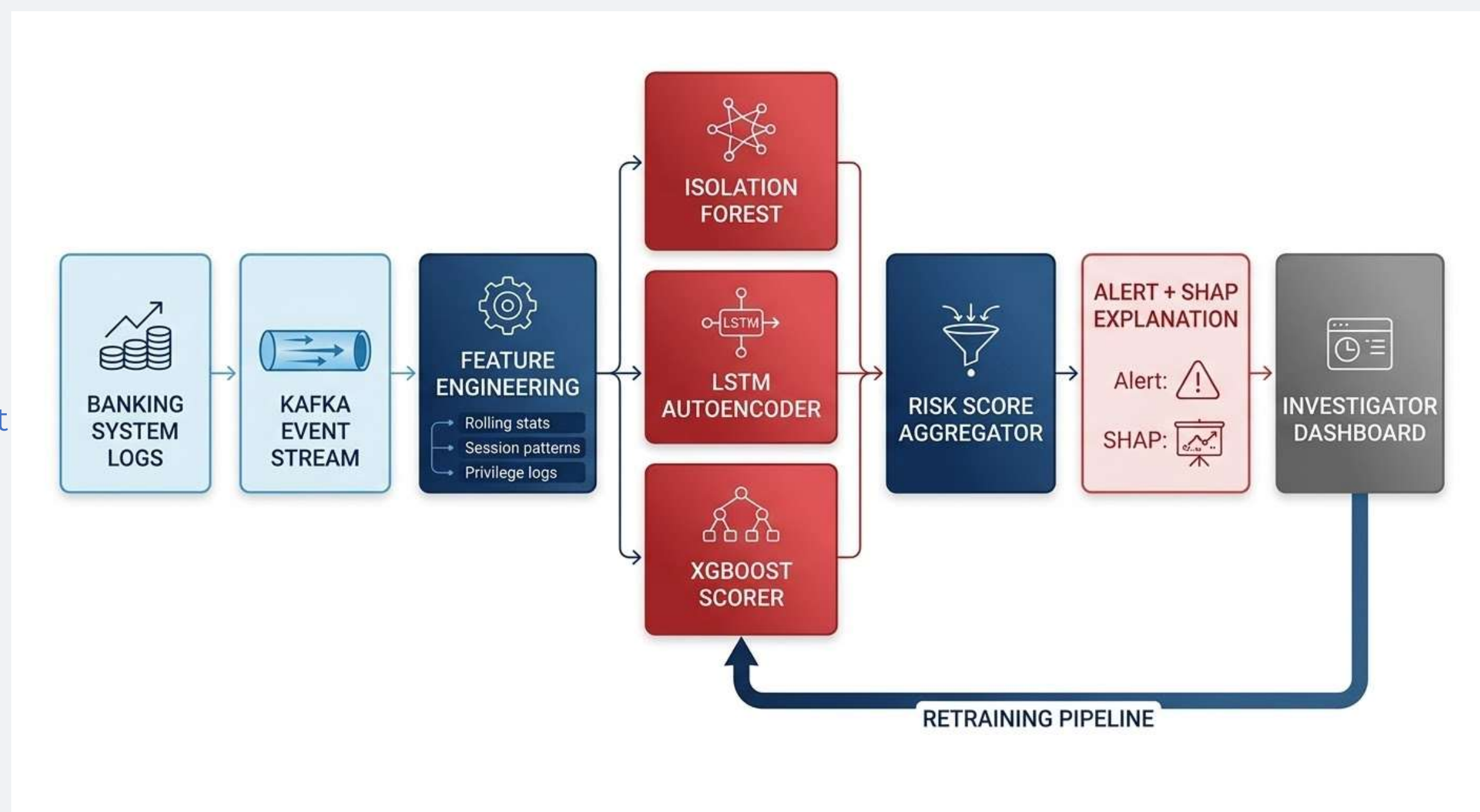
TECHNICAL APPROACH

Tech Stack:

- ML Models: Scikit-learn (Isolation Forest), PyTorch (LSTM Autoencoder), XGBoost
- Explainability: SHAP (SHapley Additive exPlanations)
- Event Processing: Kafka-inspired async event pipeline (Python asyncio)
- Backend: FastAPI with RESTful inference endpoints
- Frontend & Deployment: Streamlit

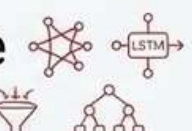
Implementation Flow:

- Banking system logs ingested via async event pipeline → per-user feature engineering (rolling 24hr/7day stats, session patterns, privilege logs)
- Three models score in parallel → Risk Score Aggregator produces final 0–100 composite score
- Threshold breach → Alert generated with SHAP waterfall explanation
- Investigator labels (TP/FP) persist to SQLite → weekly retraining pipeline triggered automatically



FEASIBILITY & VIABILITY

- All core libraries (PyTorch, Scikit-learn, XGBoost, SHAP, FastAPI, Streamlit) are production-grade and open-source
- Full prototype buildable on synthetic behavioural data — no real employee records required
- Runs entirely on standard CPU infrastructure — no specialised hardware
- End-to-end working system deliverable within hackathon timeline

Risk	Risk Level	
	Likelihood	Mitigation
Data Privacy — employee activity logs sensitive	Medium	On-premise deployment, role-based dashboard access controls
False Positive Fatigue — over-alerting causes ignored alerts	Medium	3-model consensus gate before HIGH alert fires 
Model Drift — behaviour evolves over time	Low	90-day sliding window retraining, automated weekly pipeline 
No Real Bank Data — prototype uses synthetic data	Low	Synthetic generator replicates real statistical distributions 

RETRAINING PIPELINE

IMPACT & BENEFITS

Impact:

- Reduces insider fraud detection from 12–18 months (current industry average) to real-time minutes
- Directly targets the ~30% of total Indian banking fraud losses attributable to internal actors (RBI Annual Report 2024)
- Protects millions of customers whose accounts and data are accessible to privileged bank staff
- SHAP-explained alerts fully support RBI's push for explainable AI in banking operations



Benefits:

- Economic — prevents crores in fraud write-offs, legal costs, and regulatory penalties
- Social — safeguards customer savings and personal data from privileged insider misuse
- Institutional — continuous monitoring awareness acts as a proven behavioural deterrent
- Scalable — identical system deployable across all 12 PSBs with minimal reconfiguration

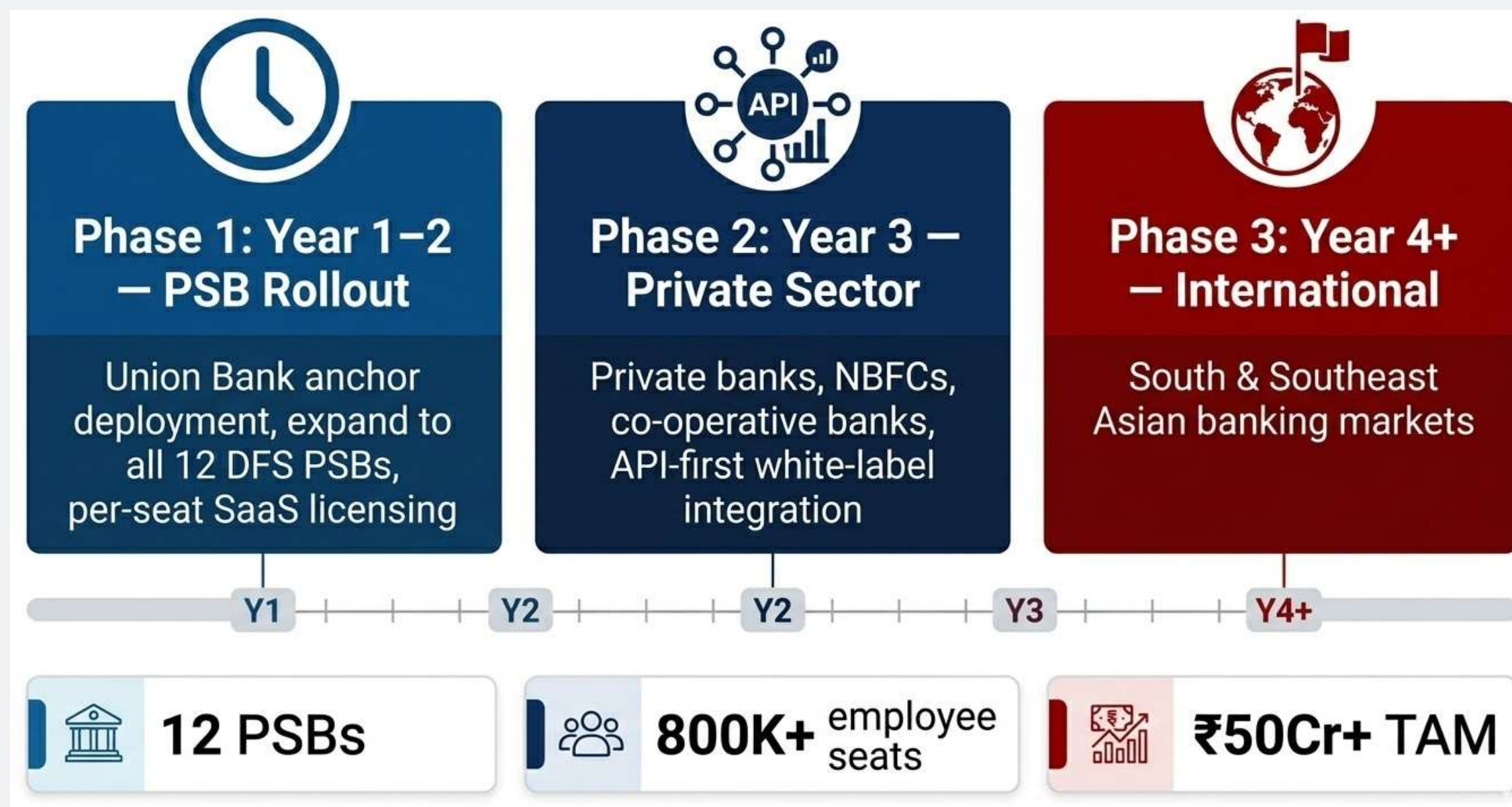
BUSINESS MODEL

Business Model:

- SaaS per-seat licensing — annual subscription priced per monitored employee, structured for PSB procurement cycles
- Union Bank as anchor customer provides immediate deployment case study and credibility for DFS-level expansion
- Clear CFO value proposition — cost of subscription directly measurable against fraud losses prevented

Commercialisation & Scalability:

- Phase 1 — 12 Public Sector Banks (~800,000 employees, largest addressable base)
- Phase 2 — Private banks, NBFCs, co-operative banks via API-first white-label integration
- Phase 3 — South/Southeast Asian markets with comparable regulatory environments
- Defensible moat — active learning model trained on Union Bank data becomes uniquely accurate for Union Bank over time, creating compounding value and high switching cost



RESEARCH & REFERENCES

- RBI Annual Report on Bank Frauds 2023–24
- ACFE Report to the Nations 2024 — Global Insider Threat Patterns
- Liu, F.T. et al. (2008) — Isolation Forest, IEEE ICDM
- Lundberg & Lee (2017) — SHAP: A Unified Approach to Interpreting Model Predictions, NeurIPS
- Hochreiter & Schmidhuber (1997) — Long Short-Term Memory, Neural Computation
- IBA Guidelines on Cyber Security for Banks (2023)
- NIST SP 800-53 — Privileged User Monitoring Controls
- **One-Page Summary** : <https://drive.google.com/file/d/1NMM9r28jNiLpud-FNChCMJrQ0bFDp-jz/view?usp=sharing>